



The macroeconomic costs of the bank tax

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ABSTRACT

In this paper, we investigate the real effects of special taxation on banks. We provide evidence that the introduction of a new fiscal levy on banks significantly impairs their performance and has an adverse impact on the real economy through the lending channel. Using micro-level data on lending relationships, we identify the credit supply shock related with a bank tax controlling for loan demand factors. We compute a firm-specific measure of firm exposure to burdened credit institutions. We find a negative impact of the tax shock on investment and output. Our results are important from a policy perspective as they shed light on the economic consequences of double taxation on banks.

1. Introduction

The global financial crisis, which sparked the most severe recession since the Great Depression, highlighted the excessive risks taken by financial institutions. The subsequent bail-outs, fiscal stimulus packages, and unprecedented monetary easing, required to prevent a global financial collapse, led policymakers to introduce new rules. These aimed at enhancing banks' resilience to shocks and ensuring a safer financial system. Key measures included higher capital and liquidity requirements (Basel, 2011). Some countries also imposed regulatory levies to mitigate banks' exposure to short-term wholesale funding or risky assets, thereby internalizing their contribution to systemic risk (Claessens et al., 2010). Furthermore, a few countries levied sectoral taxes on financial institutions to boost tax revenues (Capelle-Blancard and Havrylchyk, 2017).

The concept of additional bank taxation has reemerged as countries like Spain and Italy have implemented windfall taxes in response to increased bank profits amidst a higher interest rate environment (SandP Global, 2023; FT, 2023). Such extraordinary taxes on credit institutions raise concerns about their potential impact on financial stability. Excessive taxation can impair banks' ability to accumulate capital buffers, potentially weakening their resilience and capacity to supply credit in the economy, especially during periods of broader economic distress (ECB, 2019; IMF, 2019). Furthermore, the heterogeneous nature

of these taxes and the risk of double taxation for institutions operating across borders may contribute to fragmentation within the financial system (ECB, 2023).

The literature provides evidence that bank levies can impose significant costs on banking systems. Despite the considerable impact of such levies on credit intermediation (Boscá et al., 2019; Capelle-Blancard and Havrylchyk, 2017; Kogler, 2019; Buch et al., 2016; Sobiech et al., 2021), risk-taking (Devereux et al., 2019; Borsuk et al., 2023), and banks' financial soundness (Gropp, 2014), most empirical studies have focused on these issues from an individual bank's perspective. Consequently, the broader implications for the economy have remained largely unexplored.

In this paper, we shed light on the nexus between bank taxation and economic activity. We study how a tax on financial institutions' assets affects banks' profitability and lending behavior. We examine how the credit contraction associated with the implementation of a bank tax translates into firms' economic performance in terms of investment, sales, and employment. Additionally, we unveil the macroeconomic costs of such regulations through the lending channel.

In our study, we use the Polish bank tax, imposed on the assets of financial institutions for identification. Poland is an insightful research laboratory for several reasons. First, it is one of the few countries worldwide that has implemented a tax on bank assets. Introduced in 2016, banks in Poland have to pay a special tax calculated as 0.0366 %

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per month of the value of their assets, applicable for assets exceeding PLN 4 billion (approximately €1 billion). Second, the availability of granular supervisory bank-firm level data provides detailed information on all loans issued by banks to firms, enabling accurate identification of shifts in credit supply triggered by the new tax regime. Third, as the sixth largest economy in the European Union, Poland has a mature and well-developed banking sector with a significant presence of foreign-owned banks. Given that Polish banks operate within the European single market and their characteristics align closely with those of other European banks, we believe that our findings are applicable to other open economies.

Our research framework is developed in three interlinked stages. The first stage relies on bank-level panel data models, which enable us to investigate the impact of the introduction of a bank levy in Poland on banks' profitability and their lending to the non-financial private sector. We employ confidential unbalanced panel dataset covering all banks in Poland between 2012:Q4 and 2018:Q4. Our identification strategy relies on difference-in-difference (DiD) approach. We use unexpected introduction of the new tax on banks' assets in February 2016 as a quasi-exogenous shock. We identify banks in the treatment and control group based on the specific criteria outlined in the new tax regulation, namely the size and banks' financial standing.

The next step is to investigate whether tax reform has real consequences on firms' access to credit. Economic theory suggests that a contraction in credit supply can adversely impact corporates' operations and growth prospects, particularly if they are unable to compensate for reduced credit availability from taxed lenders with alternative funding sources, either from within or outside the banking sector (Bernanke and Gertler, 1995; Rajan and Zingales, 1998). To assess the impact of the tax on non-financial corporation (NFC) credit, we employ a loan-level dataset from Poland. This dataset aligns with the common European standard for large exposures, defining a 'large' exposure as one whose value equals or exceeds approximately 100 thousand euros. We adopt the approach suggested by Khwaja and Mian (2008), which controls for borrower riskiness and credit demand by including firm-fixed effects. Our focus is on firms that borrow from multiple banks, allowing us to attribute within-firm variations to differences in bank-specific characteristics, including asset taxation.

Finally, we explore the real effects of bank taxation from a micro-economic perspective. Our investigation focuses on how firms' exposure to taxed banks, through existing banking relationships, influences their decisions regarding investment, production, and employment. For this analysis, we integrate detailed information on banks' exposure to their counterparties with firm-level economic performance indicators from the Orbis database. We then disentangle bank-specific credit supply shocks associated with the bank tax from demand-driven shocks, following the methodologies suggested by Amiti and Weinstein (2018) and Alfaro et al. (2021). Equipped with these estimates, we construct a measure of corporate exposure to the bank tax through their banking relationships and evaluate its impact on various firm-level outcomes. Our comprehensive sample encompasses all 522 commercial and cooperative banks in Poland, which extended loans to over 17 thousand public and private firms from 2013:Q4 to 2018:Q4.

We conduct several supplementary analyses. First, we explore the heterogeneity in the impact of tax shocks on firms' economic outcomes, considering various firm-specific attributes related to their degree of financing constraints and funding diversification. Second, we utilize a dynamic treatment model—an event study approach—which helps us examine how the effects of the tax evolve over time and detect any pre-existing trends in the outcome variables that would violate the parallel trend assumption. Third, our analysis extends to the macroeconomic impact of the tax regulation, aiming to ascertain if bank-level changes influence the real economy through credit provision. We use a conditional forecasting approach, linking bank-level regression results to the framework of a new Keynesian VAR model (Lown and Morgan, 2006), thereby capturing the short-term dynamics among various endogenous

bank-specific and macroeconomic variables.

We report several novel findings. First, as expected, we observe a significant and negative association between taxation and banks' profitability. The profitability of taxed banks, measured in terms of ROA, declines on average by 0.14 pp. as a result of higher tax outlays compared to untaxed banks. Second, our regression results suggest that taxes are negatively associated with bank lending. The findings indicate that the introduction of a sector-specific tax corresponds to a 1.0 pp. decrease in the quarterly credit growth for the non-financial private sector of taxed banks, when compared to their untaxed counterparts. Our additional analysis reveals no pre-existing trend and underscores that the impact of the tax was most pronounced in the initial quarters following its implementation.

Again, using a DiD framework surrounding the implementation of the bank tax, and bank-firm level data, we find that NFCs loan growth of taxed banks contract by about 1.3 pp. in comparison to untaxed banks in response to the policy shock. Our analysis also reveals that banks reduce credit to both small and medium enterprises (SMEs) and large corporations in response to the bank tax. We also find that the bank tax adversely affects firms' economic activity. Specifically, with a one standard deviation increase in a firm's dependency on loans from taxed banks, there is an observed reduction in investments and sales by approximately 1.38 and 1.30 percentage points, respectively, compared to firms that obtain financing from untaxed banks. This effect is attributable to both SMEs and large corporations. Moreover, our findings indicate that larger firms and those with stronger financial positions are less negatively influenced by the introduction of tax shocks. Further, firms that have multiple financing sources, not solely reliant on traditional bank loans, demonstrate an enhanced capacity to manage the challenges posed by the fiscal policy shift.

Lastly, we find evidence that these effects have aggregate consequences at the country level. Our results indicate that the tax burden, which accounts for 76% of the banking sector's assets, contributes to a quarterly decline in GDP growth by an average of 0.3 pp. over a 3-year horizon. This equates to a cumulative contraction in economic growth by 3.6 pp. We conduct a series of robustness tests at each stage of our analysis, confirming that our findings remain consistent across various specifications, sample sizes (including matched samples), and estimation methods.

The novelty of our paper is threefold. First, using Poland as a research laboratory, we confirm that excessive taxation of banks can impair their profitability and credit provisioning. This aligns with previous empirical papers on bank levies, which show that financial institutions react to changes in the tax regime by increasing lending margins, decreasing deposit rates (Buch et al., 2016; Capelle-Blancard and Havrylychuk, 2017), altering the liability structure (Bremus et al., 2020), and engaging in risk-taking (Kogler, 2019; Devereux et al., 2019; Borsuk et al., 2023).

Second, our paper contributes to the literature that identifies the economic (real) effects of credit supply shocks by isolating bank-lending channel. Amiti and Weinstein (2018), Alfaro et al., 2021 and Volk (2023) disentangle credit supply shocks from demand shocks and estimate its effect on real economic outcomes. Another strand of literature focus on observable shocks to credit supply and estimate their effect on firms' economic performance. Instead of observed supply shocks related to a credit spread in Gilchrist and Zakrajšek (2012), higher capital requirements in Favara et al. (2021), sovereign debt exposure in Bottero et al. (2020), we estimate bank credit shock related to bank tax and study its real effects on employment, output, and investment.

In this context, our study is close to the work of Sobiech et al. (2021), which examines the economic effects of a bank tax on Japanese commercial banks with substantial activities in Tokyo. However, our study diverges by focusing on the impact of a tax on bank assets, which could influence banks' lending behavior differently than a tax on gross profits. Additionally, the Polish bank tax applies to all medium and large commercial banks, contrasting with the local scope of the 'Ishihara tax' in

Japan. This broader application in Poland helps mitigate potential spillover effects (Berg et al., 2021), such as borrowers seeking financing from banks outside Tokyo or from debt markets. Our study also encompasses nearly all corporate borrowers, both listed and unlisted, enabling a more comprehensive investigation into the role of financial constraints and providing novel insights into the heterogeneous responses of firms to tax shocks associated with the levy on bank assets. Overall, our results from the quasi-natural experiment with the Polish bank tax provide new findings that complement those of Sobiech et al. (2021), shedding light on the real micro-level effects of the bank tax.

Finally, this paper contributes to the literature examining the macroeconomic costs of bank regulations (Sánchez and Röhn, 2016; Kim and Mehrotra, 2018; Boar et al., 2017; Richter et al., 2019). Our work aligns with the growing body of research that employs macro-modeling approaches to assess the effects of regulatory interventions, particularly focusing on bank capital (Meeks, 2017; Conti et al., 2018; Klein and Turk-Ariss, 2022). In this context, our study makes a direct and significant contribution to the emerging literature on the macroeconomic consequences of fiscal burdens imposed on banks. While this area of research is relatively new and limited, it has revealed the adverse effects of bank taxation (Bosca et al., 2019; Restrepo, 2019; IMF, 2016). Our work complements these studies by evaluating the macroeconomic costs of a tax on bank assets, utilizing a combination of micro- and macro-modeling approach.

Our results also have broader implications for policy. While financial sector regulations aim to increase financial banks' capital and liquidity buffers to strengthen their resilience, tax policy tends to work in the opposite direction. The taxation of banks' assets decreases their resilience by compressing their profit-generating capacity and leads to lower economic activity, highlighting a fundamental inconsistency between tax and regulatory realms in public policy. Given that Poland is one of the few countries in the world that has introduced a sectoral tax on bank assets, the results of our research are particularly important in the context of other countries potentially implementing this form of taxation.

This paper is organized as follows. Section 2 provides a description of the Polish bank levy. Section 3 presents our main hypothesis and describes the main transmission channels of the tax on banks, and Section 4 presents an empirical strategy and details the data. Section 5 provides a discussion of the empirical results. Section 6 concludes the paper.

2. Bank levies at the national level: the polish case

In the aftermath of the global financial crisis, many European countries introduced regulatory levies, the goal being to internalize banks' contribution to systemic risk or to raise tax revenues (see Table 1). However, taxing models applied in Europe are not identical. Typically, the levy is established on banks' total liabilities after excluding equity and insured deposits (as in the case of Germany, the UK, Austria, Belgium, Netherlands, and Portugal). Importantly, the tax rates are relatively low, and range from values below the 20 basis points up to 40 basis points (Kogler, 2019). Moreover, several countries adopted conceptually different bank levies. In French banks, the tax is based on minimum regulatory capital, while Slovenia adopted a levy on bank assets, and Hungary a levy on loans. Finland imposed a temporary tax on risk-adjusted assets. Some countries, including Germany, have already modified, abolished or replaced the levy with mandatory contributions to the Single Resolution Fund (SRF). In addition, between 2022 and 2023, several European governments proposed a temporary windfall profits tax on banks. This came in the wake of a sharp rise in interest rates, which subsequently led to higher profits for banks (see Fig. 1).

In Poland, the bank tax was introduced much later than in most EU countries, occurring over seven years after the Lehman Brothers' bankruptcy. This delay meant the tax's anti-crisis rationale was misaligned with the country's economic situation, which was experiencing

Table 1

Selected bank taxes implemented in European countries between 2009 and 2020.

Country	Date of implementation	Tax Base	Tax rate	Exemptions
Austria	2011	Total liabilities	< EUR 20bn: 0.024 % >EUR 20bn: 0.029 %	Equity and insured deposits
Belgium	2012	Total liabilities	0.035 %	Insured deposit
Cyprus	2011	Total amount of deposits	0.15 %	Interbank deposits, deposits from foreign financial institutions
Finland*	2013	Risk-weighted assets	0.125 %	NA
France	2011	Minimum regulatory capital as determined based on RWA	0.25 %	EUR 500 million minimum own funds required
Germany	2011	Total liabilities and off-balance-sheet derivatives	0.02 % – 0.06 % 0.0003 % for off-balance-sheet derivatives.	Customer deposit and equity capital
Hungary	2010	Total assets	< HUF 50bn: 0.15 % >HUF 50bn: 0.20 %	Interbank loans
Iceland	2011	Total liabilities	0.145 %	Liabilities below ISK 50 bn
Latvia	2011	Total liabilities	0.036 %	Equity and insured deposits
Netherlands	2012	Total liabilities	0.0 % – 0.044 %	Equity and insured deposits
Poland	2016	Total assets	0.44 %	Assets below PLN 4 bn and sovereign debt
Portugal	2011	Total liabilities	0.05 %	Equity and subordinated debt
Romania	2018	Total assets	<1 % market share: 0.20 % and >1 % market share: 0.40 % with tax waivers	Cash, cash balance at central banks, sovereign debt, interbank market, NPLs
Slovakia	2012	Total liabilities	0.4 %	Equity and insured deposits
Slovenia	2011	Total assets	0.1 %	The levy is not due if the growth in lending exceeds a threshold
Sweden	2009	Total liabilities	0.036 %	Equity and insured deposits
United Kingdom	2011	Total liabilities	0.016 % short-term	Equity and insured

(continued on next page)

Table 1 (continued)

Country	Date of implementation	Tax Base	Tax rate	Exemptions
Lithuania	2023	Net interest income	liabilities 0.08 % long-term liabilities 60% on the part of NII that exceeds the average of the previous four years by 50 %	deposits with certain adjustments
Spain	2022	Net interest income and net commissions	4.8 % above a threshold of 800 million euros	
Italy	2023	Net interest income	40 % on the increase of NII	Tax capped at 0.1 % of assets

* In Finland the bank tax was collected in tax years 2013–2015. This table incorporates data and general information collected from various online sources, including the European Commission's Taxation and Customs Union Database and Reuters.

over 2.5 % GDP growth. The tax's implementation was primarily driven by the change in government after the 2015 elections. The incoming Law and Justice (PiS) party, needing additional revenue to fulfill their campaign promises, saw the bank asset tax as a way to finance these budgetary needs. Consequently, this sectoral tax became one of the new government's first significant economic policies. The rapid adoption of Poland's *Act on the Tax on Certain Financial Institutions* just 43 days post-submission to the Polish Parliament (Sejm) and its swift implementation on February 1, 2016, suggests that banks had minimal time to prepare for this new tax regime. The rapid and unexpected legislative process highlights the exogenous nature of the policy change, crucial for the context of our analysis.

The new tax in Poland targets financial institutions, including banks, insurance companies, and lending establishments. Referred to as the 'bank tax', it predominantly impacts banks, as they form the largest taxpayer group under this law. However, not every bank is subject to the tax. Institutions with assets under PLN 4 billion (around EUR 1 billion), after deducting the value of own funds and treasury securities, are

exempt. his exemption is designed to prevent undue financial burden on smaller, locally-focused institutions. Furthermore, banks participating in a recovery plan approved by the Polish Financial Supervision Authority (KNF) may also qualify for full relief. Between 2016 and 2018, approximately 76 % of banks in Poland fell within the tax's scope.

Credit institutions operating in Poland are subject to the highest effective bank tax worldwide (NBP, 2016). The tax rate is 0.0366 % of the tax base monthly, which implies an additional tax burden of almost 0.44 % annually on the taxed assets. Moreover, the new fiscal burden is not deductible, and therefore its impact on the profit of banks is substantial. The entire amount of the tax on certain financial institutions contributes to the income of the state budget. In that case, the socio-economic benefits of the regulation rest on the transfer of banks' profits to the local community through governmental revenue channels. Therefore, the Polish banking tax is purely fiscal in nature and is not established to fulfill any macroprudential objectives.

3. Hypothesis

Our framework follows the logic that the new fiscal burden on banks diminishes their profitability and affects demand for and supply of credit. Having the ability to shift a substantial portion of the cost of these new burdens onto customers, banks are expected to increase lending rates (Haskamp, 2018; Capelle-Blancard and Havrylychuk, 2017). This has a discouraging effect on credit demand. However, it is impossible for banks to fully pass the cost of the new regulation onto borrowers, and this drains their profitability. As bank profits are the main source of capital formation and constitute the first line of defense against credit losses, a higher tax burden limits a bank's lending capacity. Therefore, changes in banks' profitability caused by taxation of banks' assets impact their capital position and risk tolerance, which in turn negatively alters the credit supply (Altavilla et al., 2019). This leads us to the first hypothesis in which we presume a negative effect of the bank levy on profitability and supply of and demand for credit to the non-financial private sector:

H1. *The cost of the introduction of the banking tax on assets corresponds to reduced profitability and limited lending growth of banks*

A credit supply shock can significantly impact economic activity. This became a major topic after the global financial crisis when the bank lending channel was severely impaired, leading to a considerable decline in firms' performance in terms of investment and employment (Alfaro et al., 2021; Amity and Weinstein, 2018). A credit supply shock

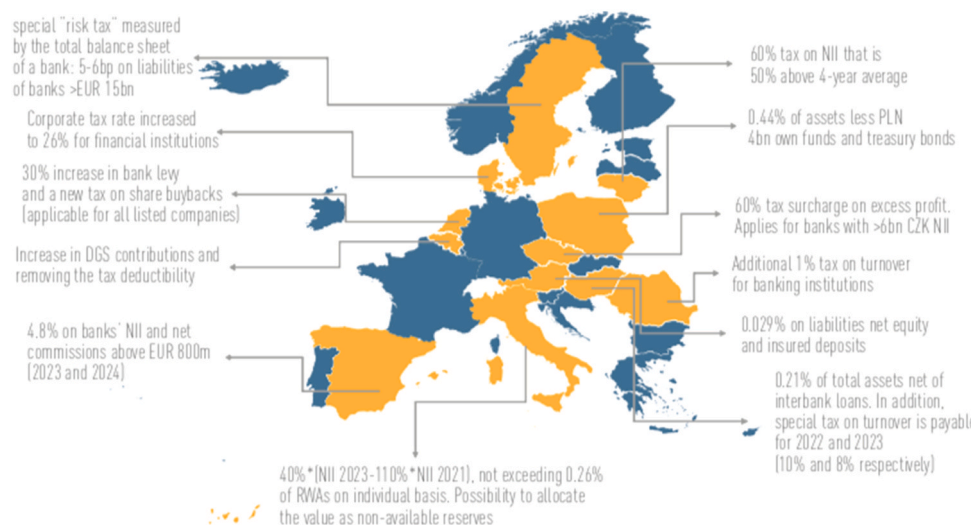


Fig. 1. Implementation of a bank-specific tax and characteristics in selected European countries.

Source: EBA, 2023

can be understood as any factor of an exogenous (e.g., monetary or macroprudential policy, new regulation) or endogenous nature (changes on banks' balance sheets) that impacts banks' lending decisions. In this context, the implementation of a Polish bank tax in February 2016, may be regarded as an exogenous shock which diminishes banks' profits and financial health. Firms, particularly those reliant on bank loans for their operational and investment activities, may face heightened financial constraints due to decrease in credit supply. This could potentially inhibit their capacity to invest in new projects, thereby influencing growth and overall firms' economic performance.

We assume that, given the fact that the Polish economy can be characterized as bank-based, bank lending conditions can have a significant impact not only on the availability of credit to firms, but also on the real firm outcomes.

H2. *The bank tax, through firms' relationships with their exposed lenders, is anticipated to adversely affect investment, production, and employment decisions within these firms*

4. Research design

4.1. Empirical strategy

In this section, we describe the identification strategy employed in the micro and macro-level setup, and the way we combine it. We assess the impact of the new bank asset tax on bank profitability, credit, and micro- and macro- economic activity. The analyses are carried out in two stages. First, we perform a bank-level analysis to assess the impact of the bank asset tax on bank profitability and lending. Second, using Polish bank-firm level data, we investigate how the shock to banks' profitability caused by the implementation of the bank tax is passed on to corporates through credit contraction in term of their economic performance.

In our supplementary analysis, we examine the dynamic effects of the tax on lending and profitability, as well as its heterogeneous impact on firms' operational outcomes. We also conduct an aggregate macro-level examination to discern the implications for real GDP growth. This sequential approach allows us to evaluate both micro- and macroeconomic consequences of the new bank tax regulation.

4.1.1. Bank level analysis

The empirical specification is designed to test whether the adoption of a special tax on financial institutions' assets leads to a reduction in banks' profitability and lending. Our identification strategy relies on a DiD approach to compare banks affected (treated) and unaffected (untreated) by the introduction of the new tax regime around its launch. The latter group was exempt from the tax due to the size criterion. In specific, the treated units are large commercial banks (19 banks), and the control entities are small commercial banks (10) and cooperative banks (493 banks). From the control group, we exclude six institutions that are undergoing recovery procedures due to their weak financial situation and the potential that their operations might be constrained by supervisory actions. We reduce the sample to the period between 2012:Q4 and 2018:Q4 to alleviate the possible problem with confounding effects. Our baseline model takes the following form:

$$Y_{it} = \beta_1 Post_t \times Treat_i + \gamma X_{it-1} + \alpha_i + \lambda_t + \epsilon_{it} \quad (1)$$

where Y_{it} denotes either the return on assets (ROA) or the quarterly growth rate of credit (*CREDIT*) in the private non-financial sector of bank i in quarter t . Our coefficient of interest is β_1 for the dummy variable $Post_t \times Treat_i$, which equals 1 for treated banks between 2016:Q1 and 2018:Q4 and 0 for all banks before 2016:Q1 and in the control group after 2016:Q1. β_1 provide us with information about the causal effect of the tax introduction on bank performance and lending behavior. We assume that affected institutions respond to changes in the fiscal policy stance in the quarter in which the new burden is in force. We expect that

the increased tax burden on banks is likely to reduce the profitability of affected entities and have a contracting effect on credit growth. In simpler terms, we expect $Post_t \times Treat_i$ to have a negative coefficient estimate in Eq. (1) for both dependent variables. We augment our model with bank- and year-quarter-fixed effects (α_i and λ_t , respectively). Bank fixed effects control for time-invariant, structural differences across banks. Time fixed effects capture unobserved time-varying factors that fluctuate over time and are common to the banking sector (e.g. macroeconomic conditions, interest rates and regulatory initiatives). Inference is done using standard errors robust in relation to heteroscedasticity and serial correlation (cluster robust). Estimation is executed using Ordinary Least Squares (OLS).

We provide several arguments that cooperative banks in Poland might serve as a reliable control group for Polish commercial banks. First, both cooperative and commercial banks operate in the same regulatory, economic, and societal environment, hence experiencing similar macroeconomic trends. Therefore, they should be similarly affected by common economic shocks or changes in the overall banking industry, apart from the impact of the bank tax. Second, both types of banks provide similar banking services, like deposit accounts, loans, and financial advisory services. Also, they serve similar categories of customers, including individuals, businesses, and public sector entities. Third, the diversity within cooperative bank reflects the heterogeneity within the Polish banking sector, making them a good control group for studying the effects of policies like the bank tax.

However, it is also essential to consider some potential differences between these groups. For example, the cooperative banks are generally smaller, less complex, have a different legal and organizational structure, and focus more on local markets compared to commercial banks. Therefore, to account for these differences, and the fact that the treatment is not randomly assigned across banks, we saturate regression with a number of bank-specific factors (X_{it-1}). We incorporate the following bank-specific control variables, namely the logarithm of total assets (*ASSETS*) to capture the economy of scale (Demirgüç-Kunt and Hui-zinga, 1999; Mirzaei, Moore and Lui, 2013), share of deposits in total liabilities (*DEPOSITS*) to capture stability and cost of funding (Berger and Bouwman, 2013), liquid assets ratio (*LIQUIDITY*) to account for the asset composition of banks and their liquidity position (Iannotta et al., 2007; Mercieca et al., 2007), the tangible equity over total assets (*CAPITAL*) to reflect the soundness of banks (Belkhir et al., 2021), the cost-to-income ratio (*Cti*) to proxy for cost inefficiency (Valverde and Fernandez, 2007; Mirzaei et al., 2013), and the ratio of the flow of new impairments over total assets (*IMPAIRMENTS*) to monitor asset quality and bank risk-taking (Albertazzi et al., 2016). We are aware that these controls may not completely capture the differences between the treated and control groups. Therefore, in our robustness section, we refine the control group and apply a matching technique to enhance the reliability of our analysis.

We use fixed values for all the control variables in the post-treatment period from 2015:Q4 (see e.g. Bottero et al., 2020; Favara et al., 2021). This strategy is employed to avoid the so-called 'bad control problems', where our results could be biased if the control variables themselves were affected by the treatment. By maintaining a static value for control variables, we strive to accurately isolate and estimate the effect of the bank tax on affected banks. In our subsequent analysis, we conduct a sensitivity test wherein we relax this constraint.¹

One key identifying assumption is that the treatment is orthogonal to the outcome variables and the entities being treated. We assume this condition is satisfied, since the treatment period is defined to start immediately after the first public announcement of the new tax regime for banks in the fourth quarter of 2015. Banks and investors did not expect this announcement, and it was instantly clear that the new

¹ We introduce time-varying explanatory variables to show the robustness of our results.

taxation system was mandatory for all large banks in Poland, with no opportunity to circumvent this new tax regulation. Therefore, the timing and mandatory nature of the policy supports the orthogonality assumption, implying that any changes observed in the outcome variables can be attributed to the treatment, i.e., the new tax regime.

The second identifying assumption of our design is the parallel-trend assumption. In other words, it must be true that, in the absence of the tax legislation, the treated group would have displayed a credit supply trend comparable to the control group. While the parallel trend assumption is fundamentally untestable due to the lack of an observable counterfactual, Section 5.3.2. presents extensive indirect evidence that supports it.

Appendix 1 presents definitions of variables and their sources. For all bank-specific variables, we winsorise observations that are below 1 or above 99 percentiles to ensure that our results are not driven by outliers.

4.1.2. Bank-firm level analysis

At this stage, we investigate whether the tax event significantly impacts firms' economic performance. The analysis is divided into three parts: estimating the impact of the bank tax on firms' credit, identifying the impact of credit supply shock on firms' economic outcomes, and evaluating the economic costs of the tax to firms. The logic behind this sequential approach is that the impact of the tax shock on lenders can only affect borrower performance if the credit supply channel is identified and indeed operative.

In the first stage, we assess the bank tax's impact on NFCs' credit supply. A common identification issue within bank lending literature is distinguishing shifts driven by demand (firms) from those driven by supply (banks). A widely-adopted strategy for addressing this issue involves controlling for demand shifts and borrower risk by including time-variant firm fixed effects in the regression models, as proposed by Khwaja and Mian (2008). Their methodology provides an intuitive framework for distinguishing simultaneous shifts in loan supply and demand, by utilizing the granularity of loan-level data. The approach is grounded on an assumption that a borrower's loan demand remains consistent across relationships with various banks within a specific timeframe. Our econometric specification takes the following form:

$$\Delta L_{fbt} = \beta_1 Post_t \times Treat_b + \gamma X_{bt-1} + \alpha_b + \lambda_{ft} + \epsilon_{fbt} \quad (2)$$

where ΔL_{fbt} is (log) growth of loans of firm f taken in bank b in year t . λ_{ft} are firm-time fixed effects that account for all the factors impacting loan growth that are consistent across banks, primarily loan demand and firm riskiness. Consequently, we rely on multi-bank relationship, i.e. firms that borrow from more than one bank as firm-time fixed effects absorb the effects on lending of firms that borrow only from one bank. A negative and statistically significant value of the coefficient β_1 would indicate the presence of the lending channel triggered by the bank tax. All the other remaining variables take the same form as those in Eq. (1). Robust standard errors are double-clustered at the bank and borrower level.

In the second stage, we temporarily set aside the tax consideration and focus on identifying the effect of the credit supply on the growth of investment, sales, and the number of employees in firms. This part of the study closely follows the approach in Alfaro et al. (2021) and Volk (2023), where the authors estimate the effect of credit supply shocks on firms' performance in Spain and Slovenia, respectively. We formulate a model utilizing borrower-specific and bank-specific dummy variables, enabling us to segregate the effects of loan demand from supply. Building on this setup, we estimate the model, where β_{bt} accounts for the supply-side factors, capturing a bank's capacity and readiness to lend:

$$\Delta L_{fbt} = \alpha_{ft} + \beta_{bt} + \epsilon_{fbt} \quad (3)$$

Next, we aggregate loan facilities and firms' financial statement data to the firm-year level and estimate the influence of loan supply shock on firm performance:

$$\Delta Y_{ft} = \gamma \bar{\beta}_{ft} + \theta controls_{ft} + \epsilon_{ft} \quad (4)$$

where ΔY_{ft} is a measure of firm performance, defined as (log) growth of (1) investment, 3 (2) sales or (3) number of employees.² $\bar{\beta}_{ft}$ is the firm-specific credit supply shock, computed as the weighted average of supply shocks deduced from the previous equation, with the loan amount serving as the weight (c_{fbt}):

$$\bar{\beta}_{ft} = \sum_b \frac{c_{fbt-1}}{\sum_b c_{fbt-1}} \hat{\beta}_{bt} \quad (5)$$

In all the estimates, we account for additional firm attributes predicted to influence firm performance. Hence, the control variables ($controls_{ft}$) encompass firm-fixed effects, firm size indicated by the log of total assets, profitability assessed by ROA, leverage gauged by equity-to-asset ratio, efficiency measured by asset-to-turnover ratio, proportion of loans on the firm's balance sheet, growth of trade credit, the number of bank relationships a firm maintains, and the estimated demand effects (α_{ft}).

In the third stage, to directly quantify how the bank tax influence firm activity, we investigate whether a firm's exposure to the tax shock has any predictive power on changes in investment, sales and employment of the NFCs in our sample. The econometric framework pursues an approach suggested by Bottero et al. (2020), Banerjee et al. (2021) and Favara et al. (2021). We start by build a firm-specific measure of exposure to the tax shock, computed by gauging the average exposure a firm f experiences through its association with its lenders b .

In Eq. (6), we define as:

$$\overline{Tax}_{ft} = \sum_b \frac{c_{fbt-1}}{\sum_b c_{fbt-1}} Tax_{fbt} \quad (6)$$

The measure ($\overline{Tax}_{ft}$ or Tax_{ft}^{AVE}) is equal to the share of taxed loan exposure held by bank b with respect to the firm f in year t (Tax_{fbt}) weighted by the share of loan that the firm has with this banks in the previous year. In other words, this measure shows how much of the firm outstanding credit granted by different lenders is subject to the new tax. Using this measure, we study how different firm-level outcomes (ΔY_{ft}) are affected by the firm-specific exposure to the bank tax shock:

$$\Delta Y_{ft} = \gamma \overline{Tax}_{ft} + \delta \bar{X}_{ft} + \theta controls_{ft} + \epsilon_{ft} \quad (7)$$

For consistency, and to capture other loan supply factors, we control for the (weighted average) of bank-specific ($\bar{X}_{ft}$) variables, borrower loan demand, constructed similar to $\overline{Tax}_{ft}$, and a battery of firm-level controls, as outlined in Eq. (4) previously. All bank- and firm- specific controls are measured at the end of the last quarter of 2015, i.e. the last quarter of the pre-tax period. In line with previous literature (e.g., Volk, 2023), we cluster the standard errors at the industry-year level.

4.2. Data and descriptive statistics

In study, we combine several databases to investigate the effect of the bank tax on banks, firms and the economy. To conduct bank-level estimations, we utilize quarterly data from all Polish commercial banks, as acquired from prudential reports. This dataset encompasses comprehensive balance sheet and income statement details, along with additional supervisory statistics. These banks are mandated to report this information under the Common Reporting Framework (COREP) and the Financial Reporting Framework (FINREP) implemented within the European Union, ensuring the reliability and comprehensiveness of our data source.

² To limit the influence of outliers, we winsorize the growth rate of investment, sales and employment at 1% level.

Data on bank tax are derived from specialized supervisory surveys conducted by the Polish Financial Supervision Authority (UKNF) from 2016 to 2018. The sample encapsulates all Polish banks that have incurred the banking tax at least once since its inauguration. The granularity of the dataset enables accurate identification of tax-paying banks, as certain financial institutions may be exempt from this obligation, for instance, due to the initiation of a recovery program.³ By the close of 2018, 19 banks had paid the bank tax, representing 76% of the assets in the Polish banking sector. The aggregate tax for a full calendar year stands at PLN 3.8 billion, equating to approximately 30% of the net profits of the banking sector.

The bank-firm level data for our study is sourced from the Prudential Reporting (so-called NB300 or 'large exposures') database,⁴ managed by the National Bank of Poland. This information is primarily used for the implementation of monetary policy and for evaluating the financial situation of banks, as well as for assessing risks within the banking sector. All banks in Poland are obligated to report exposures equal to or exceeding PLN 500 thousand (approximately EUR 100 thousand). The total exposure of the bank to a customer is calculated as the aggregate balance sheet exposure from loans and debt instruments, based on their balance value. Large exposures are reported to the National Bank of Poland on a quarterly basis, alongside other exposure-level characteristics and borrower-specific information.

Financial statement data for firms is obtained from Orbis, a global database managed by Bureau van Dijk and accessible via the Wharton Research Data Services (WRDS) platform. Orbis provides detailed and comprehensive data for both public and private companies across the globe. For our study, we specifically extract data on sales (output), investments, and the number of employees for Polish NFCs. The integration of data from NB300 and Orbis is achieved using the National Business Registry Number in Poland, known as REGON. After discarding observations with missing data, the final sample comprises 522 commercial and cooperative banks in Poland, and approximately 17,000 public and private firms covering the period from the end of 2013 to the end of 2018.

To conduct estimations at the macro level, we retrieve data from the National Bank of Poland on the Polish economy and the banking sector.

Table 2 provides the descriptive statistics of the variables used in the analysis. Panel A of Table 2 reveals that the average ROA for all banks in our sample was 0.8 %, with a standard deviation of 0.5 %. The quarterly loan growth averaged 1.4 %, with a standard deviation of 3.7 %, demonstrating robust credit growth and extensive variation in lending across banks over time. The mean values of the capital ratio (12.0 %) and credit cost ratio (0.4%) indicate robust capital positions and high-quality assets among the banks under consideration. Panel B of Table 2 report summary statistics for the bank-firm-year observations in our sample. For the typical bank-firm pair, the average annual growth to NFCs is close to 4.92 %. Firm investment, sales and employment also grew moderately during the 2013 – 2018 period. In our bank-firm database, firms, on average, have an exposure to the tax of 71% (mean of $\overline{Tax}$, with a standard deviation of 40 %). Panel C of Table 2 illustrates that the overall growth of the Polish economy was favorable for banks during this period, with the mean value for quarterly real GDP growth at 0.9 % and a standard deviation of 0.8 %.

³ Until the end of 2018, standard supervisory reports submitted by banks did not facilitate the distinction between tax costs arising from the new levy and other operational expenses.

⁴ This database is similar to the supervisory reporting (COREP 27–31) which requires SSM banks to report detailed information about their large exposures since 2014.

Table 2

Descriptive statistics of variables used in the regressions.

<i>Panel A: Bank level analysis</i>						
	N	Mean	SD	p25	Median	p75
ROA	14,368	.82	.53	.44	.77	.97
CREDIT	14,368	1.4	3.71	-1	1.39	3.24
TAX	14,368	.01	.11	0	0	0
ASSETS	14,368	18.88	1.29	18.09	18.67	19.26
LIQUIDITY	14,368	6.26	8.7	0	1.26	10.5
DEPOSITS	14,368	75.3	12.74	72.9	78.19	81.92
CAPITAL	14,368	12.09	4.17	9.07	11.35	14.15
Cti	14,368	76.06	11.73	69.06	77.01	83.82
IMPAIRMENTS	14,368	.36	.75	-.02	.11	.53
<i>Panel B: Bank-firm level analysis</i>						
NFC_CREDIT	174,202	4.92	57.32	-11.93	5.29	11.98
TAX ^{AVE}	178,770	70.88	40.49	27.08	100.00	100.00
INVESTMENT	106,597	2.71	34.87	-15.39	-5.85	15.71
SALES	106,829	2.68	30.98	-12.91	-0.18	18.98
EMPLOYMENT	28,188	3.25	20.98	-3.39	0.00	8.78
<i>Panel C: Macro-level analysis</i>						
RCREDIT	60	2.38	2.43	0.61	1.53	2.57
ROA	60	1.16	.37	.92	1.10	1.52
RGDP	60	0.89	0.80	0.63	0.98	0.9
RIR	60	4.93	3.22	3.20	4.06	5.39
REER	60	4.56	.07	4.53	4.56	4.60
RRATEAB	60	1.59	1.45	.21	1.52	2.81
EARGDP	60	.24	0.58	-.03	0.57	0.60

This table reports the summary statistics of the main variables used in the empirical analysis. The sample periods cover 2012:Q4 to 2018:Q4 for bank-level analysis after aggregation (Panel A), 2013–2018 for firm-level analysis (Panel B), and 2001:Q4 to 2018:Q4 for macro analysis (Panel C). All bank-level variables based on accounting data are winsorized at the 1st and 99th percentiles. Appendix 1 provides detailed definitions of the variables.

5. Results

5.1. Bank level analysis

In this section, we investigate the impact of the bank tax on bank performance (measured by the profitability, ROA) and bank lending (measured as the grow rate of credit to the private non-financial sector, CREDIT) with panel data regressions around the tax implementation period. We ran our baseline DiD model in Eq. (1) by comparing taxed and untaxed banks. Our coefficient of interest is β_1 for the variable ($Post_t \times Treat_i$), which equals 1 for affected banks between 2016 and 2018 and 0 for all banks before 2016 and in the control group. This coefficient provides information about the causal effect of the introduction of the new tax policy on banks.

The interaction term $Post_t \times Treat_i$ displays a negative and statistically significant coefficient at the 1% level, indicating that the introduction of the tax reform has contributed to a decline in the profitability of the treated banks, as seen in column (1) of Table 3. The influence of the tax on the banks' return on assets is not only statistically, but also economically significant, as denoted by the magnitude of the tax estimate. Consequently, due to increased tax liabilities, the profitability of the treated banks diminished by an average of 0.14 pp. compared to the untreated banks. Considering the sample mean of 0.82%, this corresponds to a decrease in profitability of about 17%.

Second, the results demonstrated in column (2) of Table 3 suggest a negative influence of the bank tax on the credit growth rates of taxed banks in comparison to their untaxed counterparts. More specifically, the coefficient estimate for the interaction term $Post_t \times Treat_i$ presents a significant negative value at the 1 % level, inferring that the new tax reform prompted a reduction in the lending capacity of taxed banks to the non-financial sector relative to their untaxed peers. The economic impact of the tax on banks' credit is substantial. The newly imposed tax liability leads to a 1 pp. decrease in the credit growth rate, representing

Table 3

Estimates from panel regressions explaining ROA and credit growth.

	(1) ROA	(2) CREDIT
TAX	-0.136*** (0.025)	-1.050*** (0.198)
ASSET	-0.790*** (0.085)	-1.599** (0.758)
LIQUIDITY	-0.001** (0.000)	0.004 (0.007)
DEPOSITS	-0.008*** (0.002)	-0.054*** (0.015)
CAPITAL	-0.006 (0.007)	0.045 (0.070)
CTI	-0.032*** (0.001)	-0.059*** (0.009)
IMPAIRMENTS	-0.301*** (0.011)	-0.326*** (0.073)
Bank FE	YES	YES
Time FE	YES	YES
R ²	0.744	0.205
Observations	14,366	14,323

This table reports OLS regression results using data for 2012:Q4 – 2018:Q4. The dependent variables represent the (log) quarterly growth rates of the non-financial loans (CREDIT) and the return on assets (ROA). TAX (Postt × Treati) is a dummy variable equal to 1 if a bank pays the tax in a particular quarter, and 0 otherwise. All regressions include bank and time-fixed effects. Appendix 1 provides detailed definitions of the variables. Robust standard errors are indicated in parentheses. *, **, and *** denote significance at the 10 %, 5 %, and 1 % levels, respectively.

71 % of the sample average and 27 % of the standard deviation in our sample.

In summary, the regression results indicate that the new tax had an immediate impact on the profitability of affected banks following its introduction, relative to unaffected banks. This suggests that banks were not able to fully pass on the cost increases associated with the new regulation to their customers, which consequently influenced their ability and willingness to extend credit to the real economy.

5.2. Bank-firm-level analysis

This section presents the results for the bank-firm level analysis. Panel A of Table 4 discusses the empirical results for the loan-level panel regression analysis based on the Eq. (2). Column 1 presents the results for the entire firm sample. Meanwhile, Columns 2 and 3 break down the results specifically for small and SMEs and large corporations, respectively. Our results indicate that, in an environment of increasing banks taxation, affected banks reduce lending to NFCs in comparison to unaffected banks. The magnitude of the $Post_t \times Treat_i$ coefficient show that the taxed banks curtail lending by about 1.3 pp. (Column 1) in comparison to untaxed banks. The specification include firm-time fixed effects to control for observed and unobserved heterogeneity that may vary over time, thus isolating credit supply from demand shocks. We can therefore conclude that the lending contraction is due to an increase in banks' taxation and is not driven by demand effects.

We also examine whether the outcomes vary based on the size of the NFCs. From a banking standpoint, SMEs and large corporations possess different risk profiles and financial needs. SMEs, often facing issues of higher revenue volatility, limited diversification, and less valuable collateral, may inherently bear a higher credit risk (Petersen and Rajan, 1994; Berger and Udell, 1995; Gertler and Gilchrist, 1994). On the other hand, large corporations, benefitting from more diverse revenue streams, more predictable demand, and established relationships with a wide range of stakeholders, typically present a lower risk profile. Our comparison of results in Columns 2 and 3 reveals that banks reduce credit to both SMEs and large corporations in response to the bank tax, suggesting that the tax impact is irrespective of firm size.

Table 4

Micro-financial implications of the bank tax – firms' borrowing from multiple banks.

<i>Panel A: Impact of the bank tax on NFCs' credit</i>			
	(1) All sample	(2) SMEs firms	(3) Large firms
TAX	-1.337** (0.676)	-1.480** (0.702)	-1.175** (0.561)
Controls	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes
Firm × Time FE	Yes	Yes	Yes
R ²	0.117	0.125	0.236
Observations	32,677	23,358	4141
<i>Panel B: Impact of credit supply on NFCs' outcomes</i>			
	Investment	Sales	Nr of employees
Credit supply	0.033** (0.016)	0.037** (0.018)	0.019* (0.011)
Controls	Yes	Yes	Yes
Time × Industry	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
R ²	0.327	0.428	0.138
Observations	13,081	13,061	2523
<i>Panel C: Impact of tax shock on NFCs' outcomes</i>			
	Investment	Sales	Nr of employees
Tax ^{AVE}	-0.034** (0.016)	-0.032** (0.015)	-0.038 (0.054)
Controls	Yes	Yes	Yes
Time × Industry	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
R ²	0.277	0.244	0.098
Observations	11,988	11,985	2268

This table presents the OLS regression results using data from 2013:Q4 to 2018:Q4. In Panel A dependent variables represent the change in the logarithm of the total loan volume to non-financial corporations at the bank-firm level (NFC_CREDIT). TAX (Postt × Treati) is a dummy variable equal to 1 if a bank pays the tax in a particular quarter, and 0 otherwise. Column (1) includes all firms in the sample, while Columns (2) and (3) specifically focus on SMEs and large corporations, respectively. Credit supply shocks are identified following the approach by Kwaja and Mian (2008), which is based on the sample of multi-bank firms using firm-time fixed effects as demand control. Panel B reports the estimated coefficients of the impact of credit supply shock on the growth of investment, sales, and the number of employees. The supply shocks are estimated with Eq. (3) and, for the purpose of estimating its impact on firm performance, are aggregated according to Eq. (5). Panel C presents estimates for the association between a firm's exposure to the tax shock and the firm's level outcome. A measure of firm-level exposure to the tax shock is computed by averaging the exposure that the firm experiences through its connection with its lenders, in line with Eq. (6). All regressions include control variables and various sets of fixed effects. Appendix 1 provides detailed definitions of the variables. Standard errors are clustered at the industry-year level and are indicated in parentheses. *, **, and *** denote significance at the 10 %, 5 %, and 1 % levels, respectively.

Next, we assess how the estimated credit supply shock, derived from Eq. (4), affects investment, sales, and employment. This evaluation precedes our main analysis, which estimates the tax shock's impact on firms' performance. Panel B of Table 4 reports our initial findings. They reveal a highly significant positive impact of the estimated loan supply on firms' investment, sales, and employee numbers. The effect is most pronounced for investments, where a one standard deviation increases in loan supply boosts investment growth by 1.15 pp., equating to 42 % of the average investment growth within the estimation period. The least notable effect relates to the number of employees, where the coefficient only reaches significance at the 10% level. Drawing on the findings from Panels A and B, we can infer that the bank tax, by influencing credit supply, may indirectly affect firms' financial outcomes. However, we cannot currently quantify this effect with precision. These insights set the stage for our subsequent analysis, which directly estimates the impact of the tax shock on firms' performance.

Panel C of Table 4 presents estimates of Eq. (7) for firm performance outcomes, with results aggregated at the firm-year level. Columns 1 and 2 suggest that firms with a higher exposure to taxed banks, on average, exhibit a reduction in investment and sales compared to those maintaining lending relationships with untaxed banks. In columns 1 and 2, the coefficients on $\overline{Tax}$ suggest that a one standard deviation increase in a firm's exposure to the tax surcharge corresponds to a decline of 1.38 pp. (i.e., 40.49×0.034) in investment growth and 1.30 pp. (i.e., 40.49×0.032) in sales growth. However, results in column 3 imply that the firms' level of employment is not significantly influenced by the companies' exposure to banks with increased asset taxation. Two potential rationales may account for this observation. First, the dataset for the number of employees has a significant number of missing observations, in contrast to the data on investments and sales, which makes it challenging to procure reliable estimates. Second, employment tends to be much stickier due to factors such as contractual obligations, labor regulations, and the indirect costs associated with workforce adjustments. These factors may render it less responsive to immediate financial shocks like the tax surcharge, especially considering the relatively short length of the sample.

In sum, our results are in line with Sobiech et al. (2021) that investigate the real financial effects of Japanese tax on banks' gross profits. Our results reveal that the bank tax on assets may have even stronger negative implications pointing out social cost of double taxation of banks.

5.3. Supplementary analysis

5.3.1. Heterogeneous impact of the tax on firms' economic outcomes

In this section, we explore the non-linearity of the impact of tax shock on firm's economic outcomes conditional on their characteristics. Extensive literature has consistently highlighted the heterogeneity of firms in their reactions to external financial shocks (Campello et al., 2010). Specifically, a firm's financial standing, capital structure, and established relationships with banks and suppliers can significantly influence the magnitude and direction of a firm's response to a credit supply shock associated with its credit taxation. To investigate this further, we introduce interaction terms between our tax measure, as defined in Eq. (6), and specific firm-level characteristics: size, profitability, leverage, efficiency, proportion of loans on the firm's balance sheet, growth of trade credit, and the number of bank relationships a firm maintains. Apart from the interaction terms, the model also incorporates each variable individually. The results are presented in Table 5.

Column 1 in Table 5 shows an interaction of a measure of firm-level exposure to the tax shocks with firm size. The positive sign of the coefficient in Panel (A) indicates that the negative impact of the tax on firm growth in investments diminishes as firm size increases. This outcome is anticipated, as larger firms typically have easier access to external financial sources besides banking loans. Consequently, their economic performance is less sensitive to changes in access to bank financing (Fazzari et al., 1988). For the remaining firm performance measures (Panel B - C), the coefficient of $\overline{Tax}$ is also positive but statistically insignificant.

In columns (2–3), we examine whether the effect of the tax is influenced by a firm's financial health by interacting our tax measure with ROA and capital ratio. According to numerous studies, low profitability and high leverage are key variables believed to be associated with firms' financial constraints (Kaplan and Zingales, 1997). The regression results suggest that constrained firms which borrow predominantly from taxed banks, experience a relatively stronger decline in the growth of investments (Panel A) and sales (Panel B). This underscores the pivotal role of a firm's financial health in moderating the impacts of external fiscal policies on its performance. Again, the results for employment growth are statistically insignificant (Panel C).

In column 4, we investigate whether the effect of the credit supply

shock, due to the bank tax, is more pronounced for firms with a significant reliance on bank financing. This is quantified by the proportion of bank loans on a firm's balance sheet from the previous year. The analysis presents distinct effects of the tax on investments and sales. Notably, firms with a greater dependence on bank financing exhibit increased sensitivity to the implications of taxing banks' assets. Conversely, results from column 5 suggest that firms with a higher engagement in trade credit are less affected by the bank tax, possibly because they might lean more on trade credit as a supplementary source when bank financing is constrained.

In column 6, we investigate the role of lending relationships in determining the influence of the bank tax on firm financial outcomes. While no discernible difference is found in the impact of the bank tax shock on sales and employment, there is a notable positive effect on investments. This suggests that firms with a greater number of credit relationships, indicative of a well-established position in the credit market, are less susceptible to credit supply shocks related to bank taxation. Their established relationships likely enable them to access a diversified pool of financing, thereby supporting their investment activities.

In summary, our findings underscore that the resilience to credit supply shocks from bank taxation is not uniform across firms; notably, smaller and financially fragile NFCs face the most pronounced challenges

5.3.2. Timing of the effects

The regression analysis in Table 3 helps quantify the overall impact of the new fiscal regulation on banks. In this subsection, we shift our focus to graphically examining the within-bank adjustments both prior to and following the policy intervention, employing an event study methodology as outlined in Baker et al. (2022) Brei et al. (2023), and Carvalho et al. (2023). This method provides visual evidence of the trends leading up to the intervention, the immediate on-impact effects, and the evolution of these effects over time. To conduct this analysis, we implement the following specification:

$$Y_{it} = \sum_{h=-H}^H \beta_h (Time_{t-h} \times Treat_i) + \gamma X_{i,t-1} + \alpha_i + \lambda_t + \epsilon_{it} \quad (8)$$

Eq. (8) extends the approach of Eq. (2) by incorporating the interaction between the treatment indicator $Treat_i$ and a series of time dummy variables $Time_{t-h}$. These dummies correspond to both pre-treatment lags and post-treatment leads, spanning up to 3 years before and after the tax regulation. This design allows us to track the quarter-by-quarter effects of the tax regulation on banks' outcome variable. To prevent perfect collinearity issues, we omit the quarter of 2015:Q4 (quarter-1) from the regression. Other regressors are identical to those in Eq. (1).

The parallel trend assumption for the treatment and control banks before the regulation is satisfied if the coefficient on the interactions involving the quarter before the event are all insignificant. The coefficient of each interaction term measures the difference between time t and time 0 in the difference in the outcome variables between treated and control banks. The sample period is the same as before, 2012:Q4–2018:Q4.

The regression coefficients, as depicted in Fig. 2, confirm that during the pre-treatment period (2012Q4 – 2015Q3), there are no statistically significant differences in ROA and lending trends between taxed and untaxed banks, whilst controlling for other important factors. This underlines the appropriateness of using smaller banks as control units for our study, given that they display parallel trends up to the reform. Further, our empirical evidence underscores that at the juncture of 2015 and 2016, the newly introduced tax policy profoundly impacted the profitability and lending activities of the taxed banks, primarily in the initial quarters following the tax implementation. Importantly, the effects of the bank tax on ROA are not significant after 3 years (as shown in the left panel), which might suggest that taxed banks reduced their asset

Table 5

Heterogeneous impact of tax shock on NFCs' outcomes.

<i>Panel A: Non-linear impact of tax shock on investments</i>						
	(1)	(2)	(3)	(4)	(5)	(6)
TAX ^{AVE}	-0.259** (0.109)	-0.063*** (0.017)	-0.137*** (0.038)	-0.074*** (0.019)	-0.033** (0.016)	-0.003 (0.024)
TAX ^{AVE} × ASSETS	0.018*** (0.006)					
TAX ^{AVE} × ROA		0.005*** (0.001)				
TAX ^{AVE} × CAPITAL			0.002*** (0.001)			
TAX ^{AVE} × LOAN_SH				-0.003*** (0.001)		
TAX ^{AVE} × TRADE_CR					0.000** (0.000)	
TAX ^{AVE} × NR_RELAT						0.013** (0.006)
Observations	11,992	11,992	11,992	11,992	11,992	11,992
R ²	0.278	0.279	0.279	0.278	0.277	0.277
<i>Panel B: Non-linear impact of tax shock on sales</i>						
TAX ^{AVE}	-0.020 (0.103)	-0.090*** (0.014)	-0.095*** (0.032)	-0.049*** (0.018)	-0.031** (0.015)	-0.025 (0.019)
TAX ^{AVE} × ASSETS	0.001 (0.006)					
TAX ^{AVE} × ROA		0.001** (0.001)				
TAX ^{AVE} × CAPITAL			0.001** (0.001)			
TAX ^{AVE} × LOAN_SH				-0.001* (0.001)		
TAX ^{AVE} × TRADE_CR					0.001** (0.000)	
TAX ^{AVE} × NR_RELAT						0.003 (0.007)
Observations	11,989	11,989	11,989	11,989	11,989	11,989
R ²	0.245	0.253	0.246	0.245	0.248	0.245
<i>Panel C: Non-linear impact of tax shock on employment</i>						
TAX ^{AVE}	-0.169 (0.262)	-0.039 (0.057)	0.035 (0.111)	-0.046 (0.056)	-0.044 (0.053)	-0.037 (0.067)
TAX ^{AVE} × ASSETS	0.007 (0.014)					
TAX ^{AVE} × ROA		-0.001 (0.003)				
TAX ^{AVE} × CAPITAL			-0.002 (0.002)			
TAX ^{AVE} × LOAN_SH				0.000 (0.003)		
TAX ^{AVE} × TRADE_CR					0.000 (0.000)	
TAX ^{AVE} × NR_RELAT						0.002 (0.007)
Observations	2266	2266	2266	2266	2266	2266
R ²	0.097	0.097	0.098	0.096	0.096	0.097
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Time × Industry	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes

This table presents the OLS regression results using data from 2013:Q4 to 2018:Q4. The dependent variables represent the growth of investment (Panel A), sales (Panel B), and the number of employees (Panel C). The estimates include only firms with multiple relations with banks. A measure of firm-level exposure to the tax shock is computed by averaging the exposure that the firm experiences through its connection with its lenders, in line with Eq. (6). All regressions include control variables (including main effects) and various sets of fixed effects. Appendix 1 provides detailed definitions of the variables. Standard errors are clustered at the industry-year level and are indicated in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

size, in line with the effects of the bank tax on credit growth (right panel), or they may have implemented other measures to offset losses. Overall, the dynamics presented in Fig. 2 align with the identifying assumption of the DiD, and argues in favor of causality between the bank tax and banks' performance.

5.3.3. Macro level impact

In examining macro-financial linkages, we analyze the aggregated consequences of the tax on the overall economic growth. Utilizing

conditional forecasting (Waggoner and Zha, 1999; Banbura et al., 2015; Andersson et al., 2010), we model an alternate scenario where the bank tax regulation is absent. We then compare the GDP growth trajectories under both the actual (with bank levy) and hypothetical (without bank levy) settings to quantify the economic effects of the new tax policy.

To bridge detailed bank-level data with macroeconomic indicators such as GDP, we employ the reduced-form Vector Autoregression model (VAR-X). The uniqueness of our approach stems from integrating bank-level regression outcomes into our VAR system. Specifically, we project

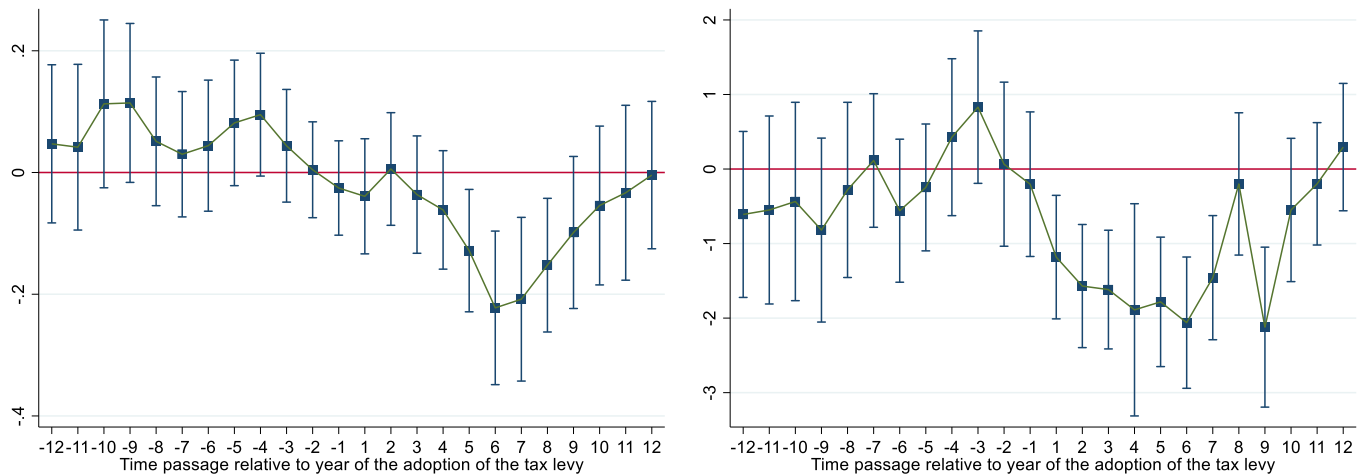


Fig. 2. The dynamic treatment effect of the tax regulation on banks' performance (left panel) and lending (right panel). The figure displays the dynamic treatment effect of the bank tax on banks' ROA and credit growth rate along with the 90 % confidence intervals. The point estimate represents the coefficient estimate of the dynamic DiD analysis of ROA and credit on relative quarterly dummies interacted with *Treat*.

the GDP growth trajectory by conditioning it on the theoretical growth paths of banking sector profitability and lending without the tax. We then compare this counterfactual to the actual GDP growth observed post-tax implementation. This method allows us to discern the tax's influence on GDP through the profitability and lending channels. The VAR-X model is constructed from aggregate data collected between 2001:Q4 and 2018:Q4, using the Cholesky decomposition. Our primary variables of interest include banking sector metrics like ROA and loan growth, GDP, among others. A detailed account of the VAR-X model is provided in [Appendix 2](#).

We construct our counterfactual using the post-tax coefficients from time-varying estimates that relate the bank tax to bank-level outcomes for each quarter, as outlined in [Eq. \(8\)](#). Such an identification scheme enables us to explore bank-level information more efficiently given the substantial degree of heterogeneity in the magnitude of banks' responses to the new taxation regime. A diagram showing research design is provided in [Fig. 3](#). This method generates 'alternative' scenario paths for ROA and credit growth by augmenting their values based on model estimates over a 12-quarter horizon. All other variables in the VAR model remain unconstrained. The impact on GDP growth, attributed to the bank levy regime, is computed by determining the disparity between scenarios where the tax is both applied and absent. Given that our time-varying DiD estimates present the difference in outcomes between the treated and control groups, we adjust these estimates by the proportion of assets held by the treated group to the total assets in the banking sector. This adjustment enables us to gauge the tax's overall impact on the profitability and lending dynamics of the entire banking industry. [Fig. 4](#) visually depicts predictions for both scenarios alongside the actual trajectory of GDP growth over a 12-quarter horizon.

The outcomes from our counterfactual analysis corroborate the insights from the bank-firm panel regression, demonstrating that fiscal measures on banks have a significant influence on the broader economy. Notably, the conditional projection of GDP growth exceeds its unconditional counterpart by an average of 0.3 percentage points over 12 quarters, leading to a cumulative decline of 3.6 percentage points in economic growth. Moreover, the different durations of the bank tax's impact on credit growth and GDP, as illustrated in [Figs. 2 and 4](#), suggest

a lag in the economy's response to fiscal changes. This is consistent with the impulse responses detailed in [Fig. 6](#), indicating a delayed yet sustained effect on GDP. Interestingly, our results somewhat align with the predictions from VAR models based on aggregate macro-financial data by the [IMF \(2016\)](#), which forecasted a 0.3–0.4 pp. reduction in GDP growth by the end of 2016 due to the asset tax introduction.⁵ Although we should interpret the magnitude of the point estimates from our macro-modeling analysis with caution, owing to data and model limitations, they do reinforce the findings from our bank-firm-level regression results. Both underscore the tangible economic costs incurred from the banking sector's double taxation

5.4. Robustness analysis

We conclude this section by conducting a series of robustness checks, which include employing alternative model specifications and conducting various sub-sample analyses at each level of our analysis. For brevity, we present results only for our main variables of interest.

5.4.1. Matched sample

One challenge inherent in our current identification strategy pertains to the comparability between banks subject to the new fiscal regulation (predominantly large commercial banks) and those not affected (primarily cooperative banks). We have addressed this concern by demonstrating pre-event parallel trends between these two groups, thus supporting the validity of our empirical approach. To further validate the robustness of our findings, we have undertaken various sub-sample analyses. These involve adjusting the composition of the control group to better align it with the treated group in terms of business model and observable characteristics.

To corroborate the evidence produced by our DiD analysis, we first refined our control sample by excluding smaller cooperative banks with assets valued below PLN 200 million (approximately EUR 50 million). This asset threshold is commonly employed by Polish authorities to distinguish between the business models of various cooperative banks. Our control sample, therefore, consists of medium and large cooperative banks, which operate more similarly to the commercial banks in our

⁵ Contrary to [IMF \(2016\)](#), which conducted its analysis based on assumptions before the tax legislation was enacted, our study leverages data available after 2016. This enables us to conduct a counterfactual analysis that was not feasible in the earlier IMF study, offering a more current and detailed evaluation of the tax's impact.

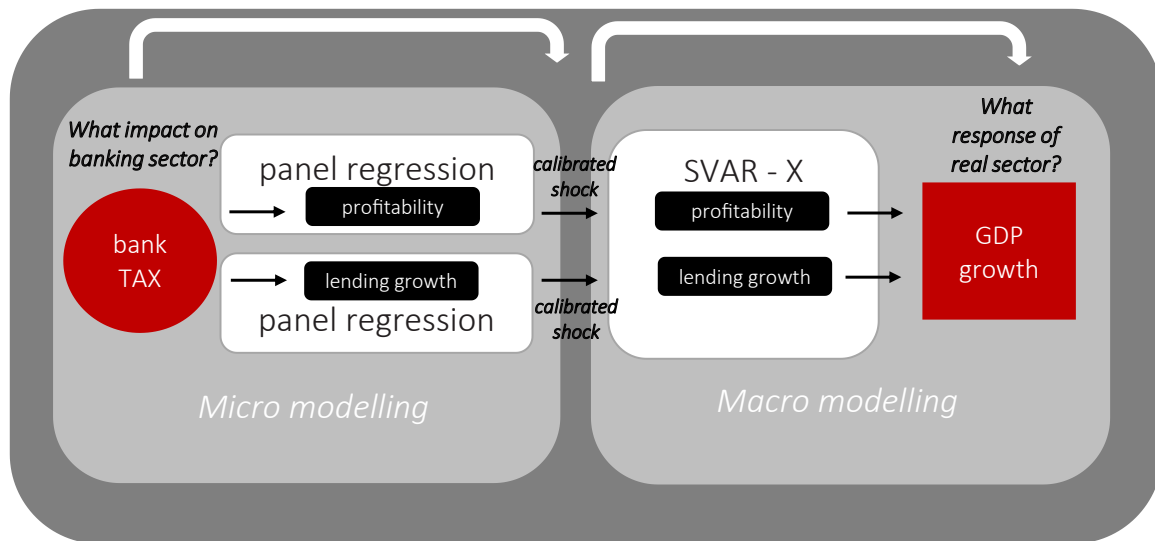


Fig. 3. Schematic illustration of the links between micro- and macro-level approach.

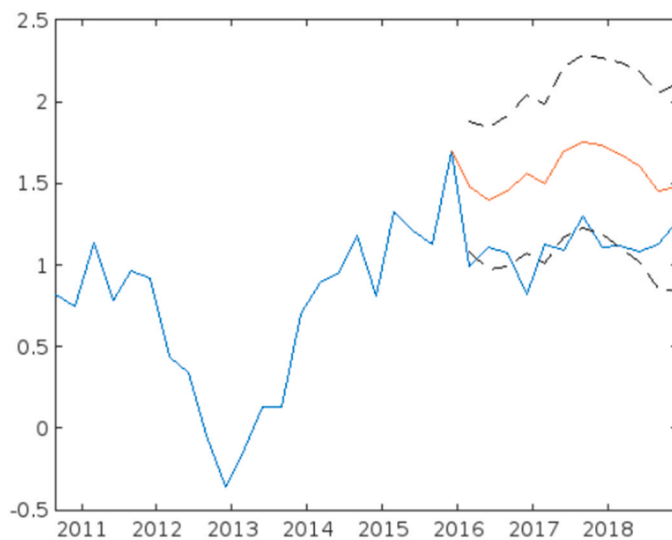


Fig. 4. Counterfactual GDP growth: without vs. with bank levy. The figure contrasts two quarterly GDP growth trajectories. The orange line represents the conditional GDP forecast based on hypothetical paths for ROA and RCREDIT (without bank levy). The blue line illustrates the actual GDP growth (with bank levy). Confidence intervals, derived from 10,000 bootstrapped replications, are indicated by grey lines at the 5- and 95-percentile levels. The GDP growth values are in percentage points.

treated group. The results presented in Panel A of Table 6 consistently show that the bank tax has a negative impact on both the ROA and credit growth of affected commercial banks when compared to these unaffected medium and large cooperative banks.

Second, we exclude cooperative banks and expand our control group to encompass branches of credit institutions (BCIs), which are predominantly not subject to the new taxation due to their size. These branches were initially not part of our regression sample because they do not hold capital independently. Their capital requirements are met by the parent companies abroad, and therefore, they were excluded from the estimation due to missing data. This adjustment allowed us to enhance our control group, which now comprises seven small commercial banks and ten BCIs. The results in Panel B of Table 6 suggest that the treated banks underperformed in terms of Return on Assets (ROA) and lending to the non-financial private sector compared to this

augmented control group. However, this analysis should be treated with caution, as many of these branches and small commercial banks often specialize in specific lending market segments and exhibit more volatile lending growth compared to larger commercial banks and cooperative banks, which were used for our baseline analysis.

In Panel C of Table 6 we employ propensity score matching (PSM) to better align our treatment and control groups before conducting our regressions. Specifically, through matching, we identify a weighted group of untaxed banks that exhibit similar characteristics to the group of taxed institutions, thereby reducing imbalance, the degree of model dependence, and the potential for bias. We choose the most straightforward matching estimator, K-nearest neighbor (NN) matching (Rosebaum and Rubin, 1983; Heckman et al., 1998). The individual from the control group is chosen as a matching partner for a treated individual who is closest in terms of the propensity score. Because our control group is larger than the treated group, we allow replacement and set $K = 3$, meaning that each treated unit is matched with up to three control units. To mitigate the risk of bad matches, which may lead to biased estimates, particularly when the closest neighbor is far away, we impose a tolerance level on the caliper at 0.05. We use logit regression to estimate the propensity score, incorporating the bank-level covariates (except for the bank assets) mentioned in Section 4.1.1. The results for the matched sample, as shown in Panel C of Table 6, are similar in both magnitude and direction to those of the unmatched sample.

Finally, in Table 7, we replicate the above analysis using micro data for the bank-firm-level relationship, in line with Eq. 2. This analysis incorporates the amended control groups, consisting of medium and large cooperative banks (Column 1), small commercial banks and branches of credit institutions (Column 2), and the matched sample (Column 3). The results based on this refined sample closely mirror those obtained with the unmatched full sample. This consistency further corroborates the robustness of our main findings, affirming the reliability of our results.

5.4.2. Other robustness tests

5.4.2.1. Bank-level robustness tests. First, we modify the specification to verify and strengthen our findings. In the baseline model, we use fixed values for all control variables in the post-treatment period, starting from 2015:Q4, to address the 'bad control' problem. Panel A of Table 8 provides a validity check, where we assess the consistency of our findings while incorporating time-varying control variables. To strengthen the reliability of the model, we incorporate pre-tax ROA in the bank

Table 6

Impact of the bank tax on ROA and credit growth – adjusting the composition of the control group in bank-level regressions.

	(1) <i>Panel A</i>	(2)	(3) <i>Panel B</i>	(4)	(5) <i>Panel C</i>	(6)
	ROA	CREDIT	ROA	CREDIT	ROA	CREDIT
TAX	-0.173*** (0.028)	-0.522** (0.218)	-0.190*** (0.071)	-1.166* (0.676)	-0.159*** (0.035)	-1.251*** (0.261)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4893	4893	863	862	1472	1470
R ²	0.754	0.295	0.724	0.137	0.685	0.288

This table presents the OLS regression results, using data from 2012:Q4 to 2018:Q4. The dependent variables represent the (log) quarterly growth rates of the non-financial loans (CREDIT) and the return on assets (ROA). TAX (Postt × Treati) is a dummy variable that equals 1 if a bank is subject to the tax in a specific quarter and 0 otherwise. Panel A refines the control sample by excluding cooperative banks with assets below PLN 200 million. Panel B excludes cooperative banks and expand control group by including branches of credit institutions. Panel C presents results for matched sample. Each regression includes bank-specific control variables, as well as bank and time-fixed effects. Detailed definitions of the variables can be found in [Appendix 1](#). Robust standard errors are indicated in parentheses, with *, **, and *** signifying significance at the 10 %, 5 %, and 1 % levels, respectively.

Table 7

Impact of the bank tax on NFCs' credit – adjusting the composition of the control group in bank-firm level regressions.

	(1) Large co-ops	(2) Commercial banks	(1) Matching
TAX	-1.887** (0.771)	-1.916*** (0.482)	-1.370*** (0.486)
Controls	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes
Firm × Time FE	Yes	Yes	Yes
R ²	0.103	0.089	0.093
Observations	28,857	22,865	22,698

This table presents the OLS regression results using data from 2013:Q4 to 2018:Q4. The dependent variable represents the change in the logarithm of the total loan volume to non-financial corporations at the bank-firm level (NFC_CREDIT). TAX (Postt × Treati) is a dummy variable that equals 1 if a bank is subject to the tax in a specific quarter and 0 otherwise. Credit supply shocks are identified following the approach by Kwaja and Mian (2008), which is based on the sample of multi-bank firms using firm-time fixed effects as demand control. Columns (1–3) present results for different compositions of the control group. Column (1) refines the control sample by excluding cooperative banks with assets below PLN 200 million. Column (2) excludes cooperative banks and expands the control group by including branches of credit institutions. Column (3) presents results for the matched sample based on PSM. All regressions include control variables and various sets of fixed effects. [Appendix 1](#) provides detailed definitions of the variables. Standard errors are clustered at the industry-year level and are indicated in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

credit equation. The results indicate that our main findings are robust to these modifications. Second, to expand the scope of our analysis, we refine our approach by including a broader set of dependent variables. Specifically, we substitute ROA with ROE, and additionally incorporate the logarithm of total assets and the Z-Score, a measure of a bank's financial stability. The results of this revised specification are presented in Panel B of [Table 8](#). Notably, we observe a negative relationship between the bank tax and ROE (Column 3), indicating that the tax adversely affects banks' capacity to generate returns for shareholders. Consistent with the credit growth model, our analysis also reveals that taxed banks reduce their asset size compared to untreated counterparts, likely as a strategy to optimize the tax base (Column 4). Importantly, the results show a decline in the Z-Score for taxed banks compared to their untaxed counterparts (Column 5), highlighting a decrease in resilience. This also indicates that the tax's adverse effect on profitability is more significant than the potential measures banks might take to increase their tax-exempt equity in their balance sheets.

5.4.2.2. Bank-firm level robustness tests. Our baseline model

Table 8

Robustness tests at the bank level.

	(1) <i>Panel A</i>	(2)	(3) <i>Panel B</i>	(4)	(5)
	ROA	CREDIT	ROE	ASSETS	Z-SCORE
TAX	-.256*** (0.245)	-1.043*** (0.215)	-1.908*** (0.241)	-0.141*** (0.006)	0.565*** (0.043)
Controls	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Observations	14,366	14,323	14,366	14,366	14,366
R ²	0.486	0.211	0.615	0.731	0.710

This table presents the OLS regression results, using data from 2012:Q4 to 2018:Q4. TAX (Postt × Treati) is a dummy variable that equals 1 if a bank is subject to the tax in a specific quarter, and 0 otherwise. Panel A accommodates changes in control variables over time. Panel B introduces new dependent variables, namely the return on equity (ROE), the logarithm of total assets (ASSETS), and Z-score ratio (Z-SCORE). Each regression includes bank-specific control variables, as well as bank and time-fixed effects. Detailed definitions of the variables can be found in [Appendix 1](#). Robust standard errors are indicated in parentheses, with *, **, and *** signifying significance at the 10 %, 5 %, and 1 % levels, respectively.

incorporates firms that engage with at least two banks. This restriction is necessary to isolate demand and supply shocks. However, this approach reduces our sample size to approximately 20 % of the total number of firms borrowing from banks, given that SMEs in Poland typically maintain a credit relationship with a single bank at any specific point in time. To address this constraint, we adopt the methodology proposed by Degryse et al. (2019), which controls for the demand side using industry-location-size-time fixed effects, enabling us to discern supply shocks even for firms borrowing solely from one bank. This approach substantially enlarges our sample size and aligns the loan portfolio structure of commercial banks more closely with that of cooperative banks, considering that the latter primarily concentrate on lending to households and smaller businesses. The result in [Table 9](#) in Panel A show that taxed banks reduce credit to NFCs strongly then untaxed banks after the tax implementation. In addition, the results in Panel B, imply that firms with greater exposure to the bank tax (indirectly through their lenders) tend to invest less (column 1) and also exhibit reduced sales (column 2). Yet, the impacts on employment fail to reach statistical significance.

5.4.2.3. Macro level robustness tests. In the macroeconomic part of our robustness analysis, we investigate the linkage between the new tax regime and economic growth using impulse response function (IRF) analysis ([Lütkepohl, 1989](#)). In our structural VAR-X model, we introduce exogenous shocks, representing the channels—profitability and

Table 9

Robustness tests at the bank-firm level.

Panel A: Impact of the bank tax on NFCs credit			
	(1) All sample	(2) SMEs firms	(3) Large firms
TAX	-1.639*** (0.256)	-1.648*** (0.253)	-1.731*** (0.292)
Controls	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes
ILST FE	Yes	Yes	Yes
R ²	0.040	0.032	0.129
Observations	103,177	89,006	14,754

Panel B: Impact of tax shock on NFCs outcomes			
	Investment	Sales	Nr of employees
Tax ^{AVE}	-0.021** (0.009)	-0.026*** (0.007)	-0.016 (0.037)
Controls	Yes	Yes	Yes
ILST FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
R ²	0.148	0.292	0.110
Observations	52,057	42,450	4875

This table presents the OLS regression results using data from 2013:Q4 to 2018:Q4. In Panel A dependent variables represent the change in the logarithm of the total loan volume to non-financial corporations at the bank-firm level (NFC_CREDIT). TAX (Postt × Treati) is a dummy variable equal to 1 if a bank pays the tax in a particular quarter, and 0 otherwise. Column (1) includes all firms in the sample, while Columns (2) and (3) specifically focus on SMEs and large corporations, respectively. Credit supply shocks are identified following the approach by Dygrese et al. (2019), which is based on the sample of both single- and multi-bank firms using industry-location-sector-time fixed effects as demand control. Panel B presents estimates for the association between a firm's exposure to the tax shock and the firm's level outcome (the growth of investment, sales, and the number of employees). A measure of firm-level exposure to the tax shock is computed by averaging the exposure that the firm experiences through its connection with its lenders, in line with Eq. (6). All regressions include control variables and various sets of fixed effects. Appendix 1 provides detailed definitions of the variables. Standard errors are clustered at the industry-year level and are indicated in parentheses. *, **, and *** denote significance at the 10 %, 5 %, and 1 % levels, respectively.

lending—through which the tax might influence the real economy. The calibration of the size of the shocks is based on panel regression models where the effect of the tax on lending and profitability is determined, as outlined in Eq. (1).

First, we investigate the effect of a hypothesized 0.14 pp. decrease in the banking sector's ROA due to the tax. Fig. 5 illustrates this channel: ROA shocks (left), which impact lending (center), and subsequently the real sector (right). In response to the drop in profitability, banks reduce their credit supply. A year after the profitability shock, quarterly credit growth is 0.3 pp. lower. This decline continues until banks' profitability returns to its original level in the following year. However, the influence of this channel on GDP, when transferred through lending, appears marginal and is statistically insignificant. This might be due to the channel's supply-driven nature. Subsequently, we adopt a more comprehensive approach, examining how the tax might influence GDP growth via banks' lending. This channel encapsulates both demand and supply-driven shocks. Guided by the outcomes from panel regressions, we simulate an approximate 1.0 pp. quarterly reduction in credit growth

following the tax's introduction. In Fig. 6, we trace the transmission channel for the credit shock (left panel) and its effect on the real economy (right panel). The tax-induced lending shock leads to a notable reduction in the quarterly output growth, with GDP declining immediately by 0.14 pp. This slowdown in growth persists for half a year before gradually rebounding over four years to its baseline. Our supplementary, yet unreported, robustness tests reaffirm the consistency of our VAR results across various model specifications, sets of variables, and estimation samples.⁶

5.5. Conclusions

In this paper, we use Poland's introduction of a tax on financial institutions' assets in 2016 as a case study to examine the economic costs of the bank tax at both micro and macro levels. Our study empirically confirms that over-taxation of banks strains their profits and, by reducing profitability, impacts lending. This, in turn, generates social costs manifested as a decrease in economic activity.

Our findings contribute to the policy debate on banking taxation. A key policy implication is that those overseeing fiscal changes in the financial industry should recognize the significance of financial intermediaries' profitability for economic growth. Notably, the introduction of these new measures should consider whether some characteristics of the taxes imposed do not entail increased uncertainty for the banking sector and lead to negative economic effects by limiting banks' ability to provide credit. Specifically, taxing banks' assets might compromise their financial health, leading to higher costs of credit intermediation, reduced loan supply, and potentially lower economic activity. While our study is centered on Poland, its implications extend to other small open economies contemplating similar tax measures. Our findings illuminate the real economic costs entailed in taxing banks, providing essential insights for policymakers in these countries.

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CRedit authorship contribution statement

Marcin Dominik: Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Anthony Saunders:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Dobromił Serwa:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Joanna Przeworska:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis.

Disclosure statement

The authors of the study declare that they have no relevant or material financial interests that relate to the research described in this paper.

⁶ Results and detailed analyses are available upon direct request to the authors.

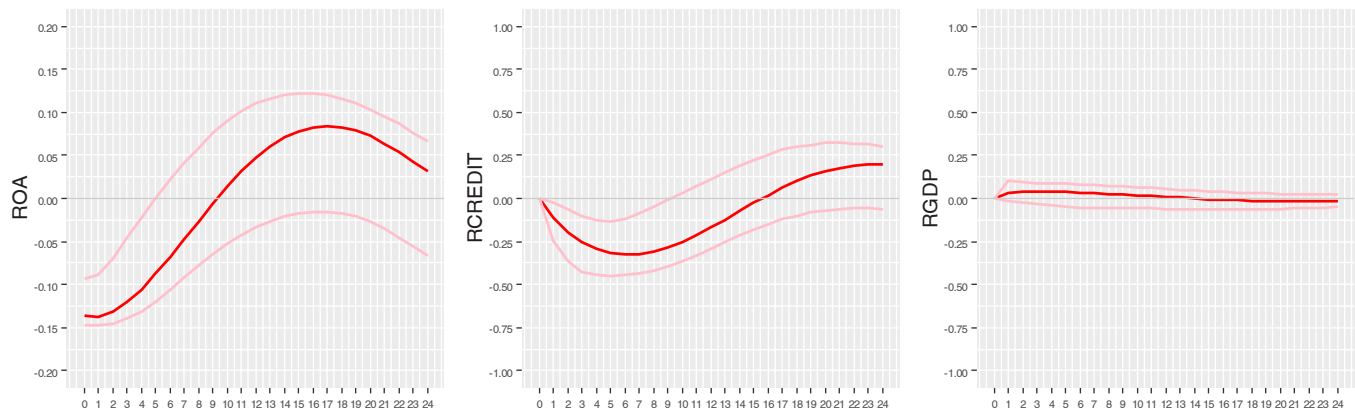


Fig. 5. The impulse responses (*ROA*, *RCREDIT*, *RGDP*) to the negative impulse from *ROA*. The panels in this figure depict the responses of selected variables to an exogenous shock to the *ROA* variable equal to -0.14 percentage points. The responses of the variables are depicted by the red lines, while the pink lines show the 5- and 95-percentile boundaries of the confidence intervals based on 10,000 bootstrapped replications. The impulse response functions show the deviation of each variable from its steady-state. *RGDP*, *RCREDIT* and *ROA* are expressed in percentage terms. The horizontal axis is in quarters.

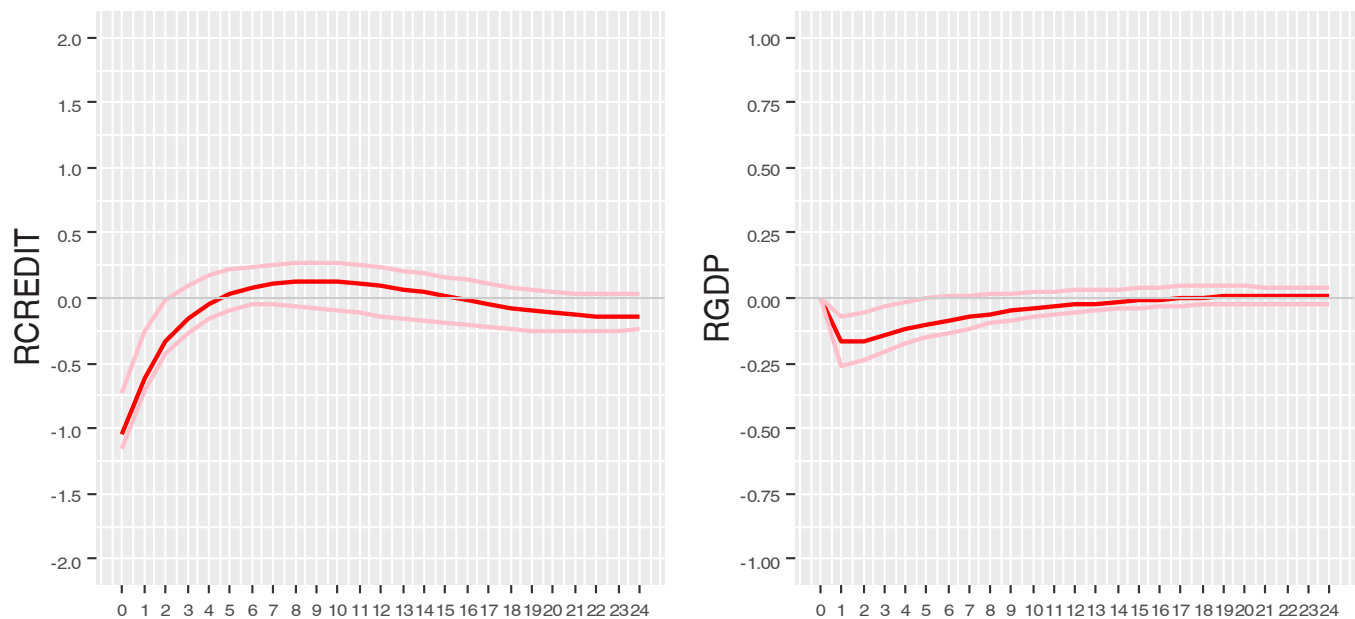


Fig. 6. The impulse responses (*RCREDIT* and *RGDP*) to the negative impulse from *RCREDIT*. The panels in this figure depict the responses of selected variables to an exogenous shock to the *RCREDIT* variable equal to -1.0 percentage points. The responses of the variables are depicted by the red lines, while the pink lines show the 5- and 95-percentile boundaries of the confidence intervals based on 10,000 bootstrapped replications. The impulse response functions show the deviation of each variable from its steady-state. *RGDP* and *RCREDIT* are expressed in percentage terms. The horizontal axis is in quarters.

Appendix 1

Table A1.1

Definition and sources of data.

Variable	Description of variables	Source
Panel A: Bank-level analysis		
TAX	Dummy variable taking the value 1 if the bank pays tax in particular quarter, and 0 otherwise	Supervisory survey
CREDIT	Credit to the private non-financial sector, log-differences, %	Supervisory reporting (FINREP/COREP)
ASSETS	Logarithm of total assets	
ROA	Net income / total assets, levels, %	
LIQUIDITY	Liquid assets / total assets, levels, %	
DEPOSITS	Deposits / total liabilities, levels, %	
CAPITAL	Equity / total assets, levels, %	
CHI	Operating cost / (operating income + amortization), levels, %	
IMPAIRMENTS	New impairments / total assets, levels, %	
Panel B: Bank-firm level analysis		
NFC_CREDIT	NFCs' credit, log-differences, Y-o-Y, %	Large-exposure database (NB300)

(continued on next page)

Table A1.1 (continued)

Variable	Description of variables	Source
TAX ^{AVE}	Firm-level average exposure on bank tax, levels, %	BvD Orbis
NR_RELATION	Nr of loan relationships a firm maintains with different banks	
INVESTMENT	Investments, log-differences, Y-o-Y, %	
SALES	Sales, log-differences, Y-o-Y, %	
EMPLOYMENT	Number of employees, log-differences, Y-o-Y %	
ASSETS	Logarithm of total assets	
ROA	Net income / total assets, levels, %	
CAPITAL	Equity / total assets, levels, %	
LOAN_SH	Loans / total liabilities, levels, %	
ASSET_TURN	Revenues / total assets, levels, %	
CURR_RATIO	Current assets / current liabilities, levels, %	
TRADE_CR	Trade credit, log-differences, Y-o-Y, %	
<i>Panel C: Macro-level analysis</i>		
RGDP	Real Gross Domestic Product of Poland, seasonally adjusted, log-differences, Q-o-Q, %	National Bank of Poland / Eurostat
RCREDIT	Real credit to the private non-financial sector, log-differences, Q-o-Q,%	
RIR	Real money market 3 M interest rate, levels, %	
REER	Real Effective Exchange Rate Index (deflated by CPI for 28 trading partners; 2010 = 100), log-levels, %	
EARGDP	Real Gross Domestic Product of Euro Area, seasonally adjusted, log-differences, %	
RRATEAB	Real money market 3 M for Euro Area, levels, %	
D2008_2009	Binary variable which equals 1 in 2008:Q4 – 2009:Q1 and 0 in the remaining quarters	

The real terms are calculated using the appropriate Consumer Price Index (for Poland and the Euro Area).

Appendix 2

The macroeconomic part of our analysis is dedicated to measuring the macro-financial consequences of introducing the bank tax. It is based on a vector autoregressive model with exogenous variables (VAR-X). The starting point is the following reduced-form VAR-X(p) model:

$$Y_t = A_0^* + A_1^* Y_{t-1} + \dots + A_p^* Y_{t-p} + D^* X_t + u_t, u_t \sim N(0, \Sigma_u) \quad (A1)$$

where Y_t is a vector of endogenous variables at time t , X_t is a vector of exogenous variables, A_0^* is a vector of constants, A_p^* is the coefficient matrix for the impact of Y_{t-p} , i.e. Y lagged p periods, on the current value Y_t , and p denotes the lag value; D is the coefficient matrix for the impact of exogenous variables X_t on Y_t , and u_t is a vector of the reduced form error terms. We employ a bootstrap method to compute standard errors based on 10,000 bootstrapped replications.

The vector Y_t contains the following endogenous variables: real output growth (*RGDP*), real credit growth (*RCREDIT*), real short-term interest rate (*RIR*), return on assets ratio of the banking sector (*ROA*), and real effective exchange rate (*REER*). These variables reflect the relationship between the financial and real sector (Goodhart, 2008; Helbling, 2011; Hristov, 2012; Mumtaz, 2015). Moreover, as a small open economy, Poland is especially vulnerable to foreign macroeconomic shocks. Therefore, the vector X_t contains the following exogenous variables: real GDP growth in the Euro Area (*EARGDP*), the real foreign short-term interest rate (*RRATE*), and a dummy variable (*D_2008_2009*), which reflects the structural break related to the global financial crisis of 2008 – 2009. The number of lags (one lag) for the VAR-X model was chosen based on the results from AIC and BIC tests. The model is estimated using aggregate macro-financial data covering the period from 2001:Q4 to 2018:Q4.

We use a structural specification of the VAR-X model to obtain economically interpretable results. We employ Cholesky decomposition, a standard and straightforward approach, to identify structural shocks in SVAR models using short-term exclusion (zero) restrictions. We base our initial ordering on the concepts adopted in literature (Berkelmans, 2005; Ciccarelli, 2015). In line with most studies, we assume that the changes in output growth (*RGDP*) have an immediate effect on all other financial (bank credit, banks' profitability, exchange rate) and macroeconomic (interest rates) variables. Changes in credit growth (*RCREDIT*) immediately affect all variables, except output growth. Following the post-crisis approach, which highlights the importance of financial intermediation, we assume that the level of interest rates (*RIR*) is dependent not only on the current stance of output growth (*RGDP*), but also on the development of credit (*RCREDIT*). The interest rates (*RIR*) have an implicit effect on banks' profitability (*ROA*) and the exchange rate (*REER*). The profitability (*ROA*) and the exchange rate (*REER*) are placed in the system at the last position, assuming it captures all available information.

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